

ASSIGNMENT 6 - Applying AI-Based Style Transfer to an Image

Comparison of two methods: pretrained CycleGAN and VGG-based Neural Style Transfer on a single landscape photograph

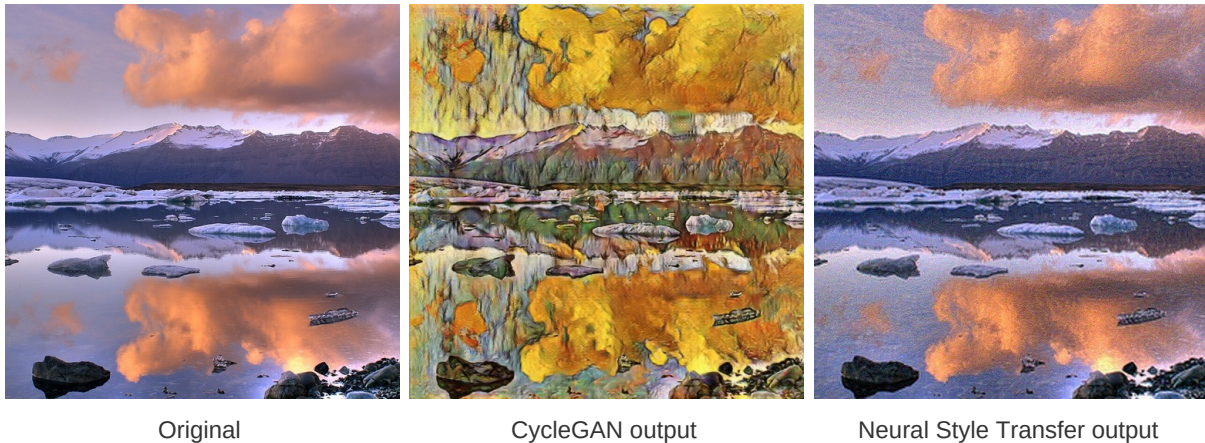


Figure 1: Original image, CycleGAN output and Neural Style Transfer output (full-size figure in comparison.png).

Method Setup

Input image: landscape photograph of Jökulsárlón glacier lagoon, Iceland (Wikimedia Commons), centre-cropped and resized to 512×512. **Method 1 - CycleGAN:** official pretrained *style_vangogh* ResNet-9 generator (junyanz/pytorch-CycleGAN-and-pix2pix released weights), photo → Van Gogh domain, input scaled to 256×256 and normalised to [-1, 1]. **Method 2 - Neural Style Transfer:** Gatys et al. optimisation using VGG-19 ImageNet features; content = relu4_2, style = relu1_1...relu5_1 Gram matrices, Adam (lr = 0.01, 120 steps, 512×512), style painting = *The Starry Night* (Van Gogh, 1889).

Written Comparison

Content preservation: Neural Style Transfer preserved the original content better — the mountain ridge, iceberg positions, rocks and reflections stay pixel-for-pixel in place, because NST only re-renders the image so that its VGG features match the style painting while the content features of the same image stay fixed. CycleGAN, by contrast, re-painted the whole scene and slightly shifted colours and small details, although the overall composition of the lagoon is still clearly readable.

Style strength: CycleGAN produced the stronger style transformation: it replaced the photographic surface entirely with thick Van Gogh-like impasto strokes and a warm yellow-orange palette, so the result looks like a painting rather than a photograph. The NST output keeps a photographic base and overlays the swirling brush texture and night-blue/gold colour scheme of *The Starry Night*, so the style is visible but softer and colour-cast.

Differences noticed: CycleGAN changes global appearance (colour, texture, lighting) with a fixed target style and needs only one forward pass, while NST changes local texture statistics and is sensitive to the content/style weight balance; NST never invents new objects, whereas CycleGAN can alter or drop small structures in the image.

Cycle consistency: CycleGAN trains on unpaired image sets using a cycle-consistency loss $\|G(F(x)) - x\|_1$ (plus the adversarial loss) that forces the two generators to be mutual inverses: a photo translated to the Van Gogh domain must be translatable back to exactly the original photo. This constraint stops the generator from collapsing to a single output and is why the original landscape structure survives the style change even though no paired photo/painting training examples were ever used.

References

Zhu, J.-Y. et al., *Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks*, ICCV 2017 (pretrained *style_vangogh* weights, official release). Gatys, L. A. et al., *A Neural Algorithm of Artistic Style*, CVPR 2016 (VGG-based NST). Images: Jökulsárlón lagoon photo and *The Starry Night* - Wikimedia Commons.